UNIVERSAL INJECTIVITY IS CHECKED AFTER A FLAT FINITELY PRESENTED BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The identification of the diagonal of a base change with a base change of the diagonal. Used at the start of the counted proof (source lines 3071-3074). Relates the two diagonals.
- The criterion for checking a property of a representable morphism along a surjective flat finitely presented cover. Invoked inside the counted proof (source lines 3075-3077) as a cross-chapter pointer. Descends surjectivity of the diagonal.
- The characterization of universal injectivity through surjectivity of the diagonal. Invoked at the end of the counted proof (source line 3078) as a cross-chapter pointer. Turns surjectivity of the diagonal into the assertion.

Source reference. Original lemma lemma-universally-injective: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2907–2907. Navigation pointer only: the counted text refers to the Stacks item lemma-universally-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $W \to \mathcal{Y}$ be surjective, flat, and locally of finite presentation where W is an algebraic space. If the base change $W \times_{\mathcal{Y}} \mathcal{X} \to W$ is universally injective, then f is universally injective.*

Proof. Observe that the diagonal Δ_g of the morphism $g : W \times_{\mathcal{Y}} \mathcal{X} \to W$ is the base change of Δ_f by $W \to \mathcal{Y}$. Hence if Δ_g is surjective, then so is Δ_f by Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering. Thus the lemma follows from the characterization (2) in Lemma lemma-universally-injective. □